CHAPTER 9

DIFFERENTIAL EQUATIONS

CHAPTER ESSENCE AT A GLANCE:
Formulation and solutions of differential equations by variable separable method, homogeneous equations, and first-order linear integrating factors.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Order and Degree Conventions:** The order of a differential equation is the order of the highest derivative present; the degree is the highest power of that derivative after clearing all fractional powers and radical expressions.

• **Separation of Variables:** Applicable when algebraic terms can be segregated so that all terms involving y appear exclusively with dy and all terms involving x appear exclusively with dx prior to direct integration.

• **Homogeneous Equations:** Differential equations where both numerator and denominator polynomials exhibit identical total algebraic degrees are resolved by substituting y equal to v times x, yielding a separable equation in v and x.

• **Linear Equations and Integrating Factors:** First-order linear differential equations are solved by multiplying the equation by an integrating factor equal to the natural exponential of the integral of the coefficient function P.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\frac{dy}{dx} = \frac{g(x)}{f(y)} \rightarrow \int f(y)\,dy = \int g(x)\,dx + C$$

General solution by variable separation method

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right) \rightarrow y = vx, \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$$

Homogeneous differential equation substitution

$$\frac{dy}{dx} + P(x)y = Q(x) \rightarrow \text{I.F.} = e^{\int P(x)\,dx}$$

Integrating factor formula for first-order linear differential equations

$$y \cdot e^{\int P(x)\,dx} = \int Q(x) \cdot e^{\int P(x)\,dx}\,dx + C$$

General solution formula for linear differential equations in y

$$\frac{dx}{dy} + P(y)x = Q(y) \rightarrow \text{I.F.} = e^{\int P(y)\,dy}$$

Linear differential equation in x and corresponding integrating factor

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Standard Form Leading Coefficient] Before identifying P and Q, the coefficient of dy by dx MUST BE 1! E.g. for $(x^2 - 1)$ dy by dx + 2xy = 1, divide entire equation by $(x^2 - 1)$ first.

• **Alert:** [Trap Alert: Particular Solution C] When initial conditions are given (e.g. y = 1 at x = 0), solve for constant C and write back the particular equation; do not stop at general solution.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define order and degree of a differential equation. Find order and degree of $(d^2y/dx^2)^3 + (dy/dx)^2 + \sin(dy/dx) + 1 = 0$. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Definitions]** 1. Order: The order of the highest derivative occurring in the differential equation. 2. Degree: The highest power (exponent) of the highest order derivative when the equation is expressed as a polynomial in all its derivatives. [1 Mark]

• **[Analysis of $(d^2y/dx^2)^3 + (dy/dx)^2 + \sin(dy/dx) + 1 = 0$]** Highest derivative is d^2y/dx^2, so ORDER = 2. The equation involves a non-polynomial term sin(dy by dx) in the first derivative, meaning it CANNOT be written as a polynomial in derivatives. Therefore, DEGREE IS NOT DEFINED. [1 Mark]

• **[Boxed Result: Order = 2; Degree is Not Defined]**

2. Verify that $y = e^{-3x}$ is a solution of $d^2y/dx^2 + dy/dx - 6y = 0$. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Verification of Solution]** Given y = e^(-3x). First derivative: dy by dx = - 3 e^(-3x). Second derivative: d^2y/dx^2 = 9 e^(-3x). [1 Mark]

• Substitute into LHS of $d^2y/dx^2 + dy/dx - 6y = 0$: LHS = 9 e^(-3x) + (- 3 e^(-3x)) - 6 (e^(-3x)) = (9 - 3 - 6) e^(-3x) = 0 · e^(-3x) = 0 = RHS. Hence y = e^(-3x) is a verified solution. [1 Mark]

• **[Boxed Result: y = e^(-3x) is a solution of $d^2y/dx^2 + dy/dx - 6y = 0$]**

3. Solve the differential equation by variable separable method: $dy/dx = (1 + y^2)/(1 + x^2)$. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Variable Separable Method]** dy by dx = $(1 + y^2)$ over $(1 + x^2)$. Separating variables: dy over $(1 + y^2)$ = dx over $(1 + x^2)$. [1 Mark]

• Integrating both sides: ∫ [dy over $(1 + y^2)$] = ∫ [dx over $(1 + x^2)$]. $\tan^{-1} y = \tan^{-1} x + C$. Alternatively: $\tan^{-1} y - \tan^{-1} x = C$ implies $\tan^{-1}$ [(y - x) over (1 + x y)] = C implies (y - x) over (1 + x y) = tan C = k. y - x = k (1 + x y). [2 Marks]

• **[Boxed Result: $\tan^{-1} y = \tan^{-1} x + C$ (or y - x = k(1 + xy))]**

4. Find the general solution of dy by dx = (1 - cos x) over (1 + cos x). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Trigonometric Simplification]** dy by dx = (1 - cos x) over (1 + cos x). Using 1 - cos x = $2 \sin^2$(x over 2) and 1 + cos x = $2 \cos^2$(x over 2): dy by dx = [$2 \sin^2$(x over 2)] over [$2 \cos^2$(x over 2)] = $\tan^2$(x over 2) = $\sec^2$(x over 2) - 1. [1.5 Marks]

• **[Integration]** dy = [$\sec^2$(x over 2) - 1] dx. ∫ dy = ∫ [$\sec^2$(x over 2) - 1] dx. y = [tan(x over 2) over (1 over 2)] - x + C = 2 tan(x over 2) - x + C. [1.5 Marks]

• **[Boxed Result: y = 2 tan(x over 2) - x + C]**

5. Find the particular solution of $dy/dx = -4x y^2$ given y = 1 when x = 0. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Variable Separation]** dy by dx = - 4x y^2. Separating variables: dy over y^2 = - 4x dx ⇒ y^{-2} dy = - 4x dx. [1 Mark]

• Integrating: ∫ y^{-2} dy = - 4 ∫ x dx implies - 1 over y = - 4 (x^2 over 2) + C = - $2x^2$ + C. 1 over y = $2x^2$ - C = $2x^2$ + k. [1 Mark]

• **[Particular Solution for y = 1 at x = 0]** 1 over 1 = $2(0)^2$ + k ⇒ k = 1. 1 over y = $2x^2$ + 1 ⇒ y = 1 over $(2x^2 + 1)$. [1 Mark]

• **[Boxed Result: Particular Solution: y = 1 over $(2x^2 + 1)$]**

6. Solve the homogeneous differential equation: $(x^2 + xy)\,dy = (x^2 + y^2)\,dx$. [Score: 5 Marks]

Answer & Key Valuation Steps:

• **[Setup & Verification]** $(x^2 + x\,y)\,dy = (x^2 + y^2)\,dx \Rightarrow \frac{dy}{dx} = \frac{x^2 + y^2}{x^2 + x\,y}$. Degree 2 in numerator and denominator, hence homogeneous. [1 Mark]

• **[Substitution y = v x]** $\frac{dy}{dx} = v + x\frac{dv}{dx}$. $v + x\frac{dv}{dx} = \frac{x^2 + v^2 x^2}{x^2 + v\,x^2} = \frac{1 + v^2}{1 + v}$. $x\frac{dv}{dx} = \frac{1 + v^2}{1 + v} - v = \frac{1 + v^2 - v - v^2}{1 + v} = \frac{1 - v}{1 + v}$. [1.5 Marks]

• **[Separation & Integration]** $\frac{1 + v}{1 - v}\,dv = \frac{dx}{x}$. $\left[-1 + \frac{2}{1 - v} \right] dv = \frac{dx}{x}$. Integrating: $-v - 2\ln|1 - v| = \ln|x| + C$. $-\frac{y}{x} - 2\ln\left|\frac{x - y}{x}\right| = \ln|x| + C \Rightarrow -\frac{y}{x} = \ln\left|\frac{(x - y)^2}{x}\right| + C$. [2.5 Marks]

• **[Boxed Result: $(x - y)^2 = C\,x\,e^{(-y/x)}$]**

7. Solve the homogeneous differential equation: $x\frac{dy}{dx} - y + x\sin\left(\frac{y}{x}\right) = 0$. [Score: 5 Marks]

Answer & Key Valuation Steps:

• **[Setup & Substitution]** $x\frac{dy}{dx} - y + x\sin\left(\frac{y}{x}\right) = 0 \Rightarrow \frac{dy}{dx} = \frac{y}{x} - \sin\left(\frac{y}{x}\right)$. Put $y = v\,x \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$. [1.5 Marks]

• $v + x\frac{dv}{dx} = v - \sin v \Rightarrow x\frac{dv}{dx} = -\sin v$. **[Separation & Integration]** $\frac{dv}{\sin v} = -\frac{dx}{x} \Rightarrow \operatorname{cosec} v\,dv = -\frac{dx}{x}$. $\ln\left|\operatorname{cosec} v - \cot v\right| = -\ln|x| + \ln C = \ln\left|\frac{C}{x}\right|$. $\operatorname{cosec}\left(\frac{y}{x}\right) - \cot\left(\frac{y}{x}\right) = \frac{C}{x}$. Using $\frac{1 - \cos\left(\frac{y}{x}\right)}{\sin\left(\frac{y}{x}\right)} = \tan\left(\frac{y}{2x}\right)$: $x\tan\left(\frac{y}{2x}\right) = C$. [3.5 Marks]

• **[Boxed Result: $x\tan\left(\frac{y}{2x}\right) = C$ (or $x\left[\operatorname{cosec}\left(\frac{y}{x}\right) - \cot\left(\frac{y}{x}\right)\right] = C$)]**